

Total No. of Questions : 8]

SEAT No. :

PC-2639

[Total No. of Pages : 2

[6354]-788

B.E. (AI&DS)

INFORMATION RETRIEVAL

(2019 Pattern) (Semester - VII) (417524 B) (Elective IV)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagram must be drawn wherever necessary.
- 3) Assume suitable data, if necessary.

Q1) a) Define and explain Probabilistic Retrieval. [6]

b) What is the binary independent retrieval model? [6]

c) Explain types of language models? [6]

OR

Q2) a) List and explain the challenges in Natural language processing [6]

b) State and explain different types of Bayesian networks? [6]

c) Explain with suitable example different levels of NLP [6]

Q3) a) What is the difference between clustering and classification? Can clustering be used for classification purposes? [6]

b) What are different types of clustering algorithm? Explain any one of them. [5]

c) Explain in detail about naïve Bayes algorithm and its application in text classification [6]

OR

P.T.O.

Q4) a) What is K nearest neighbor classifier with an illustration in information retrieval? [6]

b) Explain Mixture of Gaussians Model. [5]

c) Solve Agglomerative hierarchical clustering for single link with example. [6]

Q5) a) Factors that influence a webpage's PageRank score [6]

b) Discuss the factors that influence a webpage's PageRank score. Discuss the challenges involve in web search engine [5]

c) Define Python library is used for web crawling? [6]

OR

Q6) a) Define Python library is used for web crawling? [6]

b) What are the Main challenges posed by Web? [5]

c) Explain the components of focused web crawlers? [6]

Q7) a) Explain multimedia IR models? [9]

b) What are the four phases of recommender system? [9]

OR

Q8) a) What are the different evaluation metric for Recommender system. [9]

b) What is the difference between information retrieval and recommender system? [9]

